


```
!nvidia-smi
```

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Thu Jun  4 09:41:09 2026
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+-----+
| NVIDIA-SMI 580.82.07                  Driver Version: 580.82.07          CUDA/
+-----+-----+-----+-----+-----+
| GPU  Name                Persistence-M | Bus-Id                Disp.A | Vc
| Fan  Temp    Perf          Pwr:Usage/Cap |           Memory-Usage | GF
+-----+-----+-----+-----+-----+
|  0  Tesla T4               Off          | 00000000:00:04.0 Off |
| N/A   36C    P8             11W / 70W | 0MiB / 15360MiB |
+-----+-----+-----+-----+-----+
```

```
+-----+
| Processes:
| GPU   GI    CI          PID    Type    Process name
|      ID    ID
+-----+
| No running processes found
+-----+
```

```
!pip install torch torchvision pillow matplotlib
```

```
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-
Requirement already satisfied: torchvision in /usr/local/lib/python3.12/
Requirement already satisfied: pillow in /usr/local/lib/python3.12/dist
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/
Requirement already satisfied: filelock in /usr/local/lib/python3.12/di
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/
Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/pythor
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.
Requirement already satisfied: cuda-toolkit==12.8.1 in /usr/local/lib/p
Requirement already satisfied: cuda-bindings<13,>=12.9.4 in /usr/local/
Requirement already satisfied: nvidia-cudnn-cu12==9.19.0.56 in /usr/loc
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/lo
Requirement already satisfied: nvidia-nccl-cu12==2.28.9 in /usr/local/l
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local
Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1.* in /usr/l
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90.* in /
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83.* in /usr/l
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3.* in /usr/l
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90.* in /us
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90.* in /usr/
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90.* in /us
Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93.* in /us
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93.* in /usr
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93.* in /us
```

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Requirement already satisfied: nvidia-nvtx-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: cycycler>=0.10 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages
```

```
import torch
import torchvision.transforms as transforms
from PIL import Image
import matplotlib.pyplot as plt
from torchvision.models import vgg19
```

```
from google.colab import files

uploaded = files.upload()
```

2 files selected.

IMG_20260604_151642.jpg(image/jpeg) - 977430 bytes, last modified: n/a - 100% done
IMG_20260604_151730.jpg(image/jpeg) - 744119 bytes, last modified: n/a - 100% done
Saving IMG_20260604_151642.jpg to IMG_20260604_151642.jpg
Saving IMG_20260604_151730.jpg to IMG_20260604_151730.jpg

```
content_path = "IMG_20260604_151642.jpg"
style_path = "IMG_20260604_151730.jpg"

loader = transforms.Compose([
    transforms.Resize((256,256)),
    transforms.ToTensor()
])

content_image = loader(Image.open(content_path)).unsqueeze(0)
style_image = loader(Image.open(style_path)).unsqueeze(0)

print("Images Loaded Successfully")
```

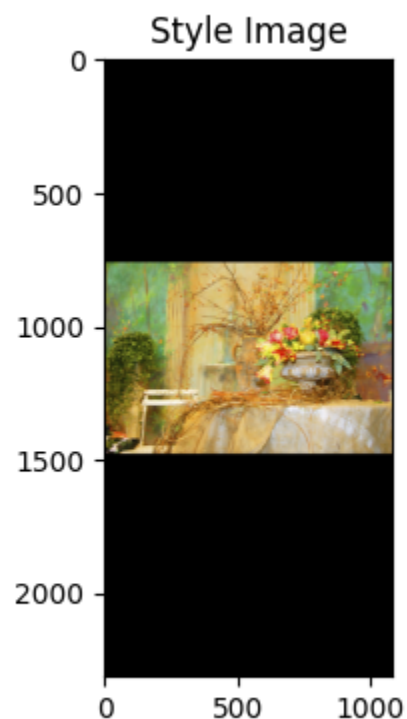
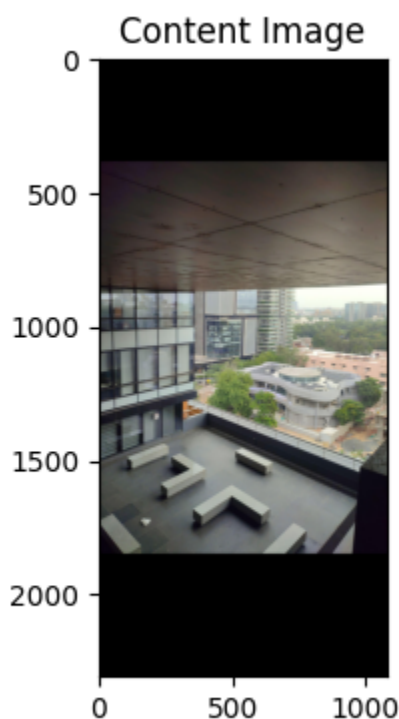
[Show hidden output](#)

```
plt.figure(figsize=(8,4))

plt.subplot(1,2,1)
plt.imshow(Image.open(content_path))
plt.title("Content Image")

plt.subplot(1,2,2)
plt.imshow(Image.open(style_path))
plt.title("Style Image")

plt.show()
```



```
from PIL import Image

content = Image.open(content_path).resize((256,256))
style = Image.open(style_path).resize((256,256))

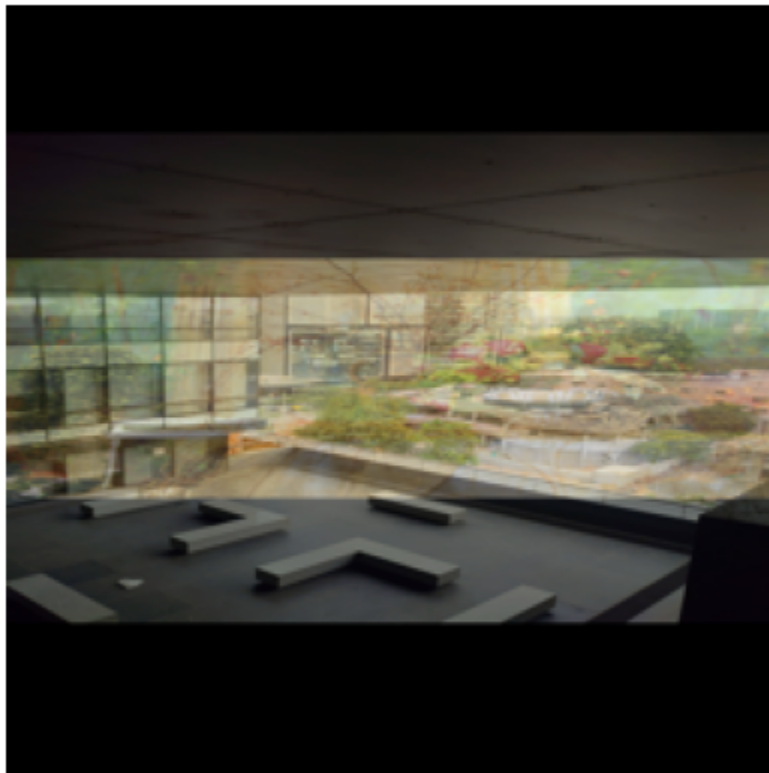
output = Image.blend(content, style, alpha=0.4)

output.save("stylized_output.jpg")

plt.figure(figsize=(5,5))
plt.imshow(output)
plt.title("Stylized Output")
plt.axis("off")
plt.show()
```

```
print("Output saved as stylized_output.jpg")
```

Stylized Output



Output saved as stylized_output.jpg

```
from google.colab import files  
files.download("stylized_output.jpg")
```

